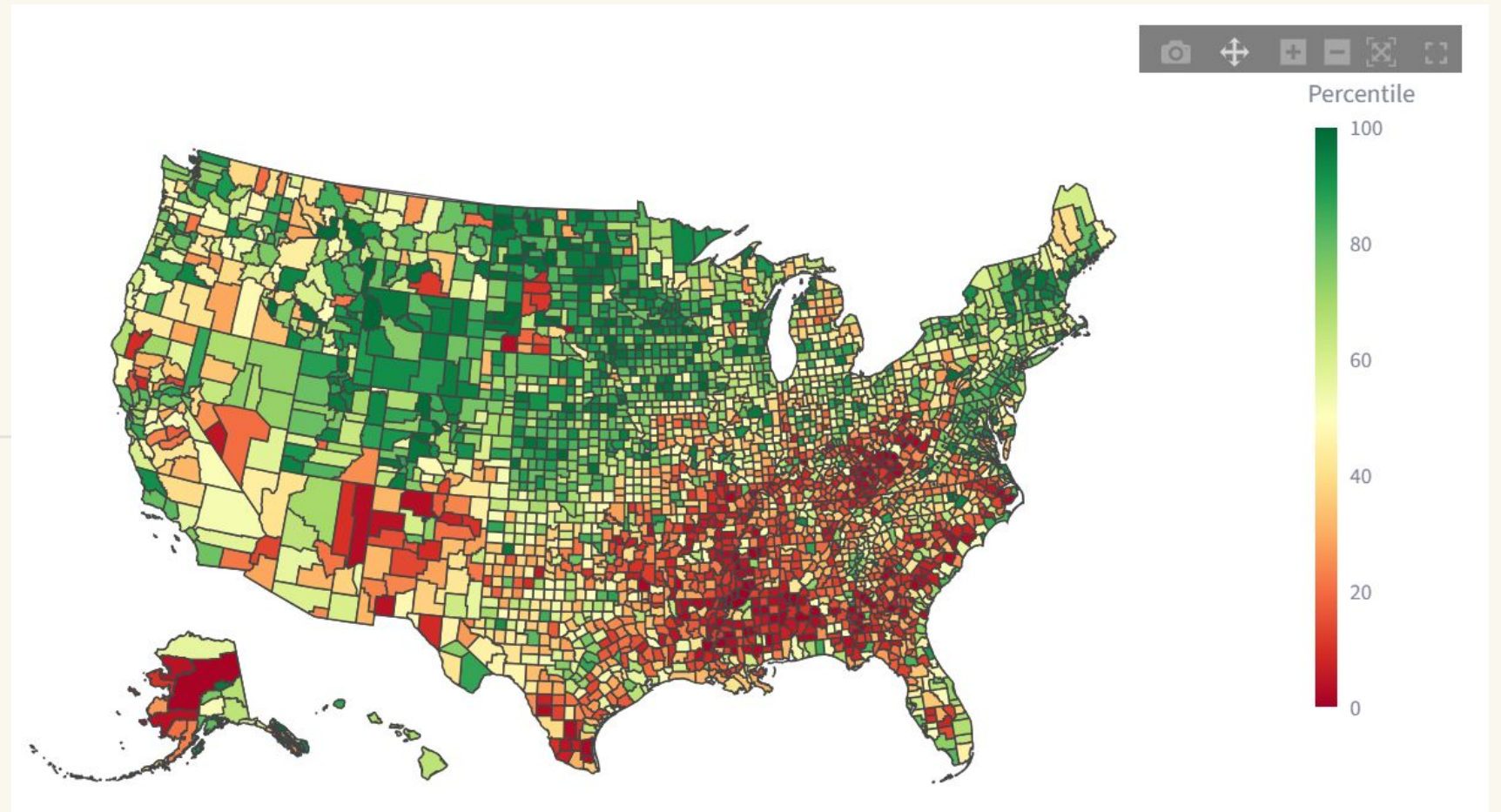


# Customizable County-Relocation Ranking

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# Summary

- Main question: What counties look historically strongest for relocation once we account for multiple sustainability categories and user preferences?
- Unit of analysis: U.S. county
- Output: personalized ranking + heatmap
- A county-level scoring system that turns many sustainability and quality-of-life indicators into an interactive map where users can choose their own priorities.

# Problem Setup: Relocation reasons are subjective

The project turns subjective priorities into a transparent county-level score.

1

## Stakeholders

People considering relocation, planners, and anyone comparing counties across livability and sustainability dimensions.

2

## Decision context

Users do not all care about the same things, so the final ranking should change when their priorities change.

3

## Checks

Score coverage across counties, interpretability, ranking stability, and external comparison with migration rates.

### THE EIGHT CATEGORY SCORES



Food, Water & Amenities



Air Pollution & Climate Risk



Health & Quality of Life



Financial Sustainability



Housing



Mobility & Infrastructure



Security



Social Capital & Community

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**Value add: not one fixed “best places” list, but a customizable map of trade-offs.**

# Data story: One county-level table built from many sources

## DATA STRUCTURE

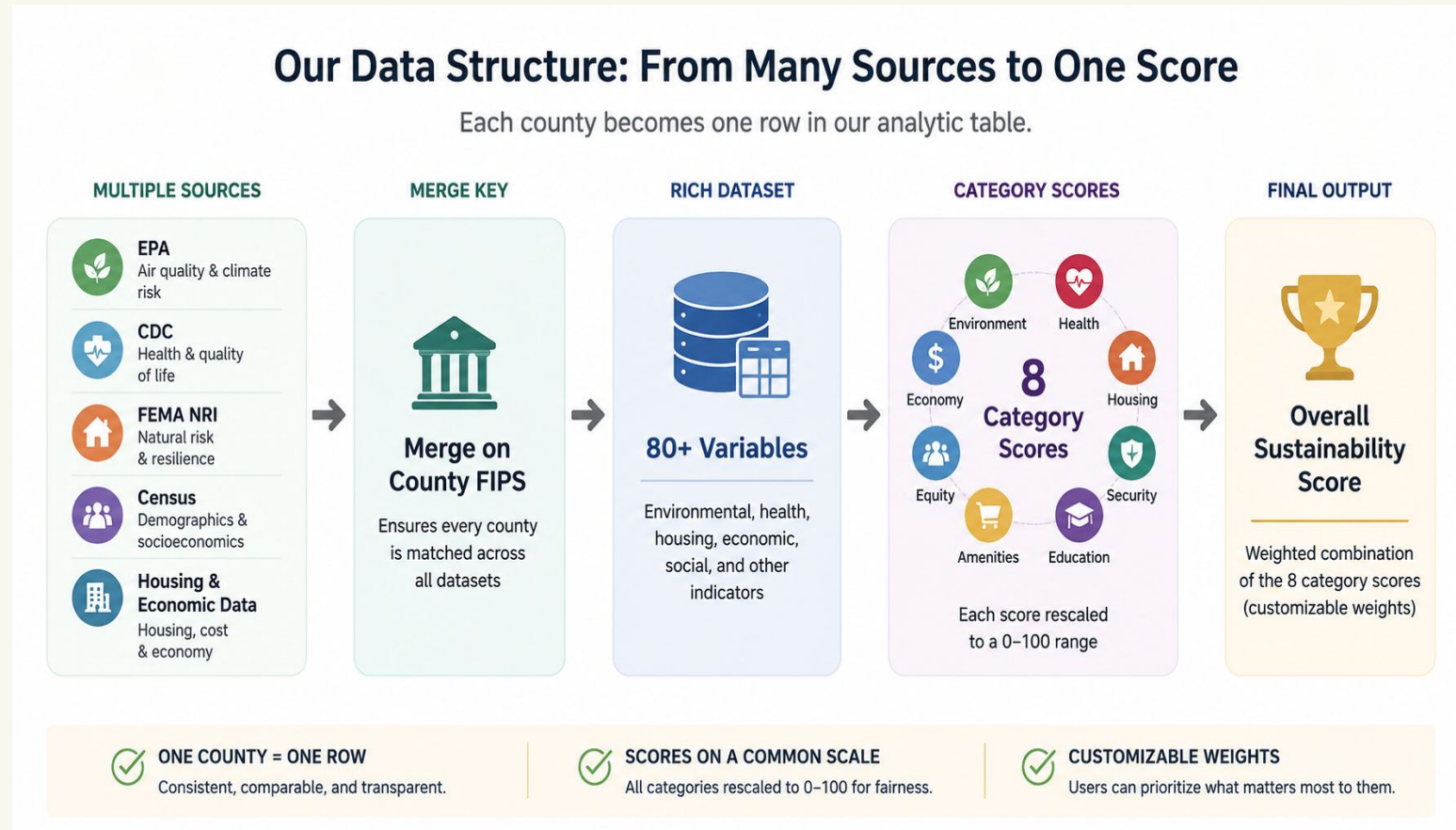
- County FIPS
- 0–100 category scores
- 80+ variables

## CLEANING

- ✓ Missing data handled
- ✓ Duplicate counties removed
- ✓ Variables standardized

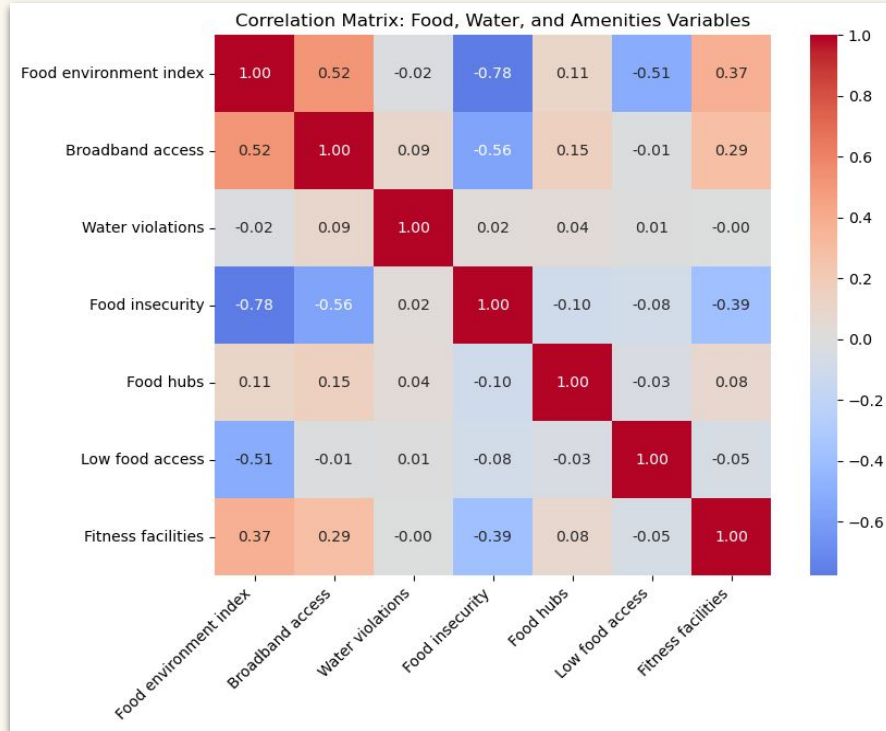
## LIMITATIONS

- Missing coverage
- Migration used for validation

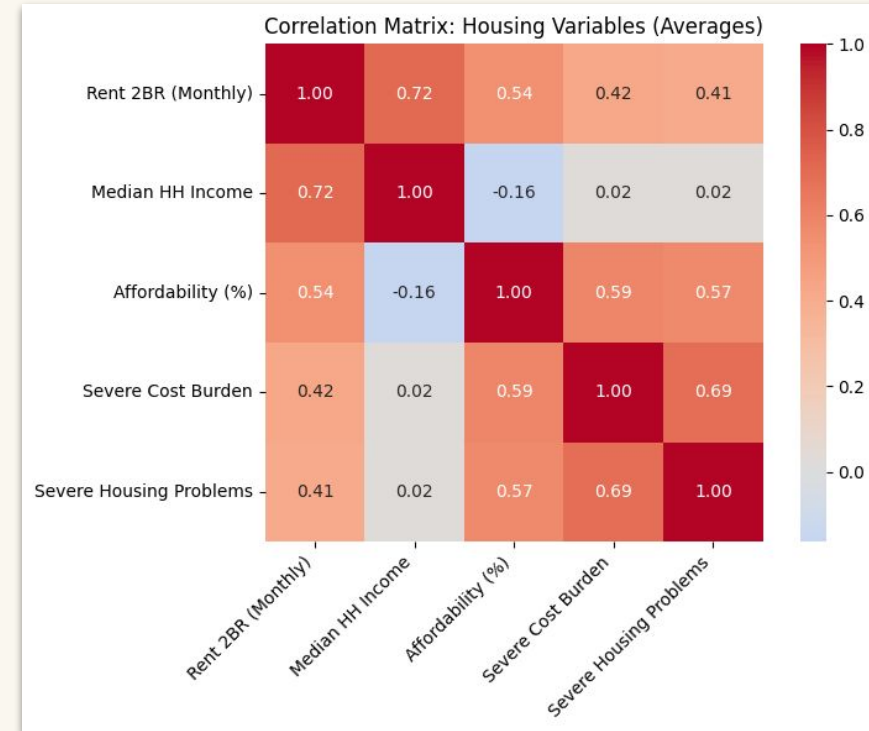


# EDA: correlations reveal structure inside categories

Why not just average all the variables?



Food, Water & Amenities



Housing

## EDA takeaway

Many variables were highly correlated, so PCA removed redundant information while preserving the most important patterns.

# Feature Engineering + Scoring Model

Goal: turn many county-level variables into **8 category scores** that users can **weigh** based on their own **priorities**

## CATEGORY SCORING PIPELINE

### Select Variables

Selected relevant variables for each category using EDA, domain knowledge, and data availability



### Adjust Directions

Adjusted directions so that higher values always mean “better”



### PCA

Used components from reduced variables to summarize each category into one score



### Rescale Scores

Rescaled scores to between 0–100 to use in app

## HOW PCA IS USED

- Reduced the number of variables within each category
- Retained the components that captured most of the variation
- Selected the number of components separately for each category to balance dimensionality reduction and information retention

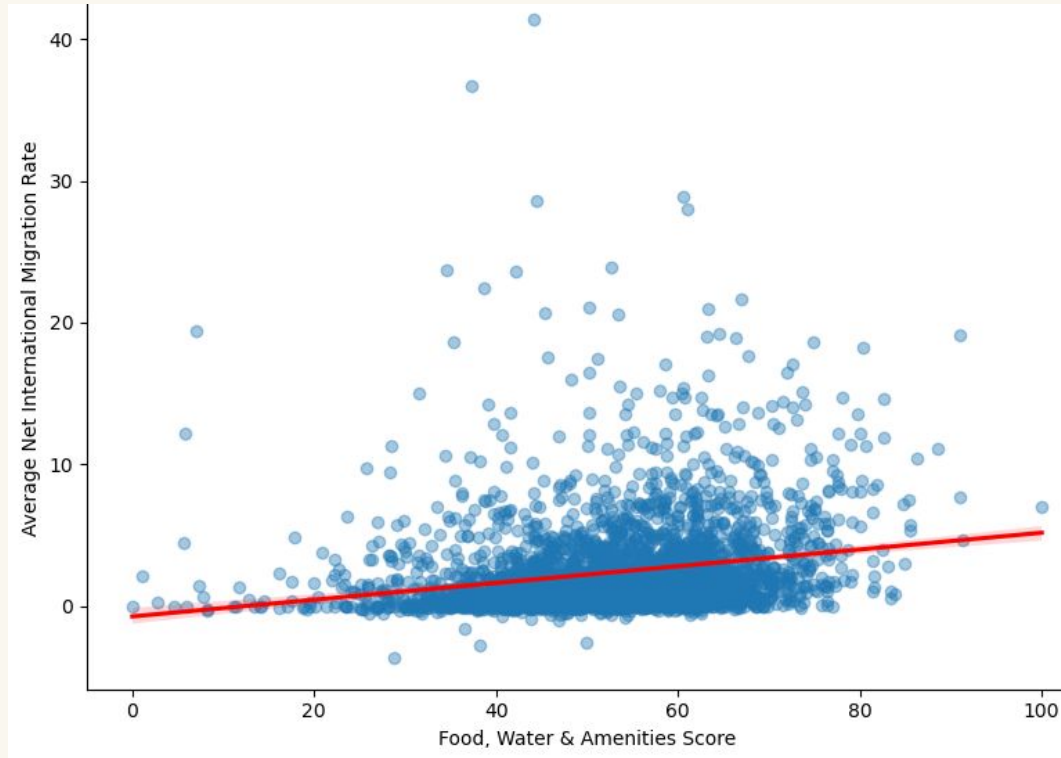
## WHY PCA FIT OUR GOAL

- Condensed correlated county-level variables into fewer uncorrelated components
- Created standardized category scores for comparing counties
- Simplified the data structure used in our customizable relocation-ranking app

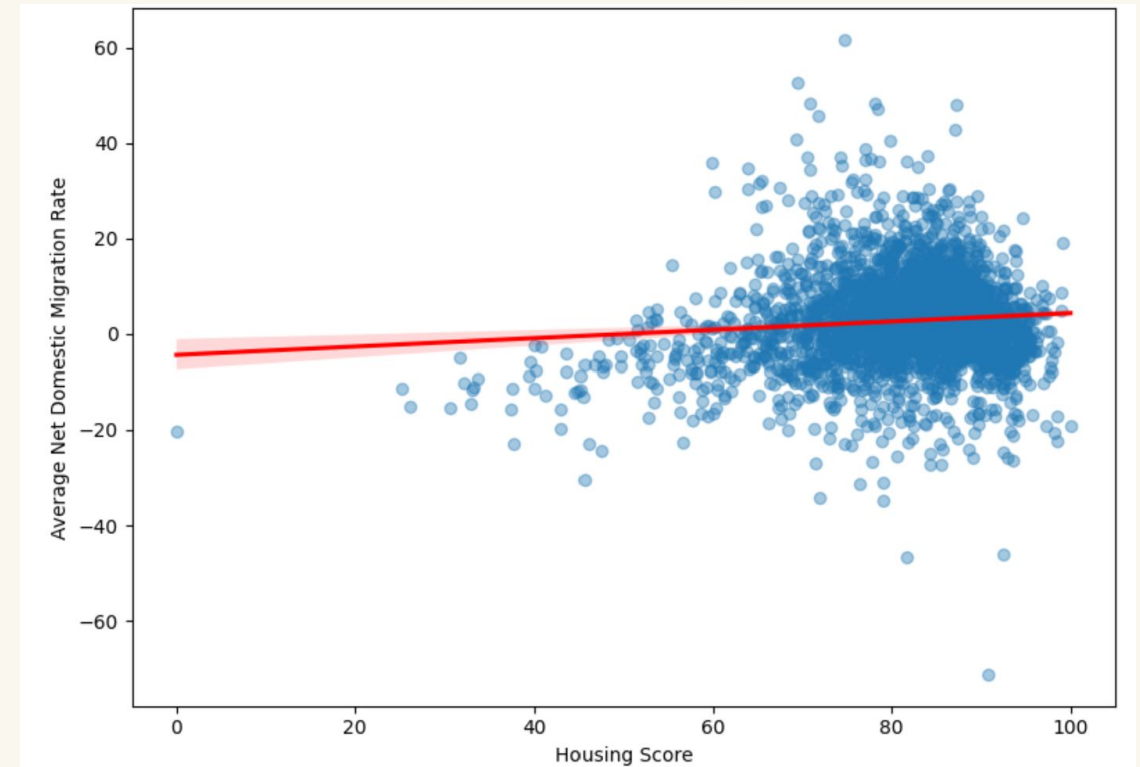
# Validation: Comparing our scores to migration rates as a sanity check

A single category score is **not expected to fully explain migration**, but comparing it with migration data helps us check whether the score **behaves reasonably**.

Food, Water & Amenities Score vs. International Migration Rate



Housing Score vs. Domestic Migration Rate



## Observations

- Several category scores show **weak or modest links** to migration.
- Often stronger for international migration than domestic migration.

# Limitations and Next Steps

The tool is a strong first version, but future work should improve robustness, validation, and interpretability.

## 1 Data Limitations

County scores depend on available county-level data, year coverage, and proxy variables. Some measures may not fully capture lived experience.

## 2 Migration is Imperfect

Weak migration correlations do not mean the rankings are wrong. Relocation reflects jobs, family, housing costs, policy, and personal constraints — not just quality of life.

## 3 Next Steps

Run sensitivity analyses on category weights, compare multiple weighting scenarios, update with newer data, and validate against external benchmarks. Future versions could also adapt the framework to compare cities or regions globally using internationally available indicators.

## Takeaway

Our tool is not meant to produce one universal “best county” list. Its value is that it creates a transparent, customizable ranking system that lets users prioritize what matters to them and compare trade-offs visually.

**Not just one “best county” list — a tool for comparing what matters.**